

Supplementary Material

From Laboratory to Field: A Leakage-Controlled Cross-Domain Evaluation Protocol for Plant Disease Recognition

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This supplement carries the complete metric tables behind the compact versions in the main paper, the per-seed values every aggregate is computed from, and the per-run diagnostic figures. Nothing here is a new result: it is the same artifacts at full width.

All values come from `experiments/ica26/metrics/*.json` via `scripts/ica26_build_tables.py` and `scripts/ica26_sig`. The corpus is bound to dataset lock `83449f6013691855`. This version covers 9 of 18 planned runs across seeds 42, 1337.

1 Dataset Provenance and Leakage Control

Table 1. Dataset provenance and leakage control. Perceptual-hash Hamming threshold 6; all 16 near pairs human-adjudicated as clearly different.

Quantity	Value
PlantDoc acquired (source)	2578
PlantDoc previous effective	2564
PlantDoc Core	2561
PlantDoc non-reviewed unchanged	2554
PlantDoc retained reviewed	7
PlantDoc excluded reviewed	17
Cross-dataset exact duplicates (acquired)	0
Cross-dataset near duplicates (acquired)	16
Cross-dataset exact duplicates (Core)	0
Cross-dataset near duplicates (Core)	16
Unresolved leakage pairs	0
Shared cross-domain classes	21
Cross-domain evaluation images	1951

2 Complete Performance Tables

Table 2. In-domain test performance, mean $\pm$ sample standard deviation over independent training seeds (42, 1337, 2026). Macro averages are computed over the classes present in the evaluation split; PlantDoc Core’s arthropod-pest class has zero test images and is reported separately rather than averaged in as zero.

Model	Dataset	Seeds	N	Accuracy	Macro-F1	Weighted-F1	Balanced Acc.	Macro Prec.	Macro
ResNet-50	PlantVillage	1	10709	0.9925	0.9899	0.9925	0.9898	0.9903	0.9899
EfficientNet-B0	PlantVillage	1	10709	0.9968	0.9956	0.9968	0.9954	0.9958	0.9956
MobileNetV3-Small	PlantVillage	2	10709	0.9950 $\pm$ 0.0001	0.9929 $\pm$ 0.0002	0.9950 $\pm$ 0.0001	0.9917 $\pm$ 0.0001	0.9942 $\pm$ 0.0005	0.9917
ResNet-50	PlantDoc Core	1	225	0.6933	0.6994	0.7157	0.6813	0.7585	0.6813
EfficientNet-B0	PlantDoc Core	2	225	0.6689 $\pm$ 0.0157	0.6795 $\pm$ 0.0187	0.6965 $\pm$ 0.0202	0.6583 $\pm$ 0.0118	0.7466 $\pm$ 0.0073	0.6583
MobileNetV3-Small	PlantDoc Core	2	225	0.5578 $\pm$ 0.0031	0.5893 $\pm$ 0.0075	0.5931 $\pm$ 0.0061	0.5654 $\pm$ 0.0024	0.6589 $\pm$ 0.0182	0.5654

Table 3. Cross-domain generalisation, mean $\pm$ sample standard deviation over three seeds: PlantVillage-trained models evaluated on the frozen shared-class PlantDoc Core subset. Metrics are computed only over the 21 shared classes; unmapped PlantDoc images are excluded from the evaluation set rather than counted as errors. Retained probability mass is the share of the source model’s belief falling inside the shared space before renormalisation.

Model	Dataset	Seeds	N	Accuracy	Macro-F1	Weighted-F1	Balanced Acc.	Macro
ResNet-50	PlantVillage $\rightarrow$ PlantDoc Core	1	1951	0.2778	0.2414	0.2663	0.2590	0.3483
EfficientNet-B0	PlantVillage $\rightarrow$ PlantDoc Core	1	1951	0.2742	0.2359	0.2618	0.2602	0.3285
MobileNetV3-Small	PlantVillage $\rightarrow$ PlantDoc Core	2	1951	0.2263 ± 0.0221	0.1873 ± 0.0154	0.2018 ± 0.0226	0.2203 ± 0.0115	0.3232

Table 4. Domain-shift degradation, mean $\pm$ sample standard deviation over three seeds. Drops are computed within each seed and then averaged, preserving the pairing between a checkpoint and its own cross-domain evaluation. In-domain is the PlantVillage test split over 38 classes; cross-domain is the PlantDoc Core shared-class subset over 21 classes. The two label spaces differ, so the drop combines domain shift with the change of task difficulty and should be read as a paired trend across models rather than as an isolated quantity.

Model	Seeds	Accuracy (in-domain)	Accuracy (cross-domain)	Accuracy abs. drop	Accuracy rel. drop %	Macro-F1 (in-domain)
ResNet-50	1	0.9925	0.2778	0.7147	72.01	0.9899
EfficientNet-B0	1	0.9968	0.2742	0.7226	72.49	0.9956
MobileNetV3-Small	2	0.9950 ± 0.0001	0.2263 ± 0.0221	0.7687 ± 0.0222	77.26 ± 2.22	0.9929 ± 0.0002

Table 5. Model efficiency, mean $\pm$ sample standard deviation over three seeds. Epoch counts vary between seeds because early stopping fires at different points, so training time is reported alongside seconds per epoch. Latency and throughput are measured after warm-up on the accelerator reported in the run metadata.

Model	Dataset	Seeds	Params (M)	Epochs	Train time (min)	s / epoch	Latency b1 (ms)	Throughput (img/s)	Peak mem (GB)
ResNet-50	PlantVillage	1	23.59	11.0	120.8	658.8	8.43	219.8	946.0
EfficientNet-B0	PlantVillage	1	4.06	14.0	132.1	566.2	9.31	275.0	923.0
MobileNetV3-Small	PlantVillage	2	1.56 ± 0.0	15.0 ± 0.0	34.4 ± 1.3	137.7 ± 5.1	6.60 ± 0.07	1294.4 ± 26.3	215.0
ResNet-50	PlantDoc Core	1	23.57	10.0	12.1	72.7	8.44	216.1	125.0
EfficientNet-B0	PlantDoc Core	2	4.04 ± 0.0	9.0 ± 0.0	10.1 ± 2.4	67.5 ± 16.0	9.75 ± 0.24	158.1 ± 165.2	117.0
MobileNetV3-Small	PlantDoc Core	2	1.55 ± 0.0	13.0 ± 2.8	2.7 ± 0.6	12.4 ± 0.1	6.48 ± 0.45	1272.1 ± 2.1	214.0

Table 6. Confidence calibration and selective prediction, mean $\pm$ sample standard deviation over three seeds. Temperature is fitted on the held-out validation split only and applied unchanged to test and cross-domain logits; it is never refitted on evaluation data. AUROC is the area under the risk-coverage curve; lower is better.

Model	Dataset	Setting	Seeds	NLL	Brier	ECE	MCE	AUROC
ResNet-50	PlantVillage	in-domain	1	0.1158	0.0198	0.0903	0.2958	0.00
ResNet-50	PlantVillage	in-domain (T-scaled)	1	0.0304	0.0112	0.0094	0.3912	0.00
ResNet-50	PlantVillage	cross-domain	1	2.9677	0.9991	0.3038	0.4924	0.63
ResNet-50	PlantVillage	cross-domain (T-scaled)	1	3.6646	1.1098	0.4305	0.6488	0.63
EfficientNet-B0	PlantVillage	in-domain	1	0.1146	0.0150	0.0985	0.4481	0.00
EfficientNet-B0	PlantVillage	in-domain (T-scaled)	1	0.0213	0.0051	0.0093	0.5666	0.00
EfficientNet-B0	PlantVillage	cross-domain	1	2.8580	0.9309	0.2390	0.4116	0.58
EfficientNet-B0	PlantVillage	cross-domain (T-scaled)	1	3.4337	1.0303	0.3778	0.5604	0.58
MobileNetV3-Small	PlantVillage	in-domain	2	0.1207 ± 0.0044	0.0192 ± 0.0013	0.0987 ± 0.0036	0.6252 ± 0.1568	0.00
MobileNetV3-Small	PlantVillage	in-domain (T-scaled)	2	0.0269 ± 0.0017	0.0083 ± 0.0003	0.0092 ± 0.0004	0.3552 ± 0.0009	0.00
MobileNetV3-Small	PlantVillage	cross-domain	2	2.7766 ± 0.0224	0.9259 ± 0.0029	0.1478 ± 0.0172	0.4550 ± 0.0116	0.67
MobileNetV3-Small	PlantVillage	cross-domain (T-scaled)	2	3.1017 ± 0.0244	1.0118 ± 0.0072	0.2900 ± 0.0157	0.5480 ± 0.0324	0.67
ResNet-50	PlantDoc Core	in-domain	1	1.2001	0.4685	0.1071	0.3183	0.10
ResNet-50	PlantDoc Core	in-domain (T-scaled)	1	1.1822	0.4582	0.0618	0.1824	0.10
EfficientNet-B0	PlantDoc Core	in-domain	2	1.1906 ± 0.0031	0.4822 ± 0.0023	0.1376 ± 0.0251	0.5626 ± 0.4521	0.10
EfficientNet-B0	PlantDoc Core	in-domain (T-scaled)	2	1.1289 ± 0.0014	0.4598 ± 0.0022	0.0564 ± 0.0111	0.2519 ± 0.1082	0.10
MobileNetV3-Small	PlantDoc Core	in-domain	2	1.5351 ± 0.0130	0.6056 ± 0.0061	0.1417 ± 0.0102	0.2995 ± 0.0279	0.24
MobileNetV3-Small	PlantDoc Core	in-domain (T-scaled)	2	1.5005 ± 0.0092	0.5946 ± 0.0000	0.1029 ± 0.0216	0.2399 ± 0.0032	0.24

Table 7. Backbone ranking by macro-F1 under each evaluation setting, per seed. A setting is STABLE when all seeds agree on the ordering. This is the direct test of whether the in-domain ranking predicts the cross-domain ranking, or whether an apparent reordering is within seed noise.

Setting	Seed Ranking (macro-F1; best first)	ResNet-50 macro-F1
PlantVillage in-domain	42 EfficientNet-B0 > MobileNetV3-Small > ResNet-50	0.9899
PlantVillage in-domain	all STABLE: 1/1 seeds give EfficientNet-B0 > MobileNetV3-Small > ResNet-50	0.9899
PlantVillage → PlantDoc Core	42 ResNet-50 > EfficientNet-B0 > MobileNetV3-Small	0.2414
PlantVillage → PlantDoc Core	all STABLE: 1/1 seeds give ResNet-50 > EfficientNet-B0 > MobileNetV3-Small	0.2414
PlantDoc Core in-domain	42 ResNet-50 > EfficientNet-B0 > MobileNetV3-Small	0.6994
PlantDoc Core in-domain	all STABLE: 1/1 seeds give ResNet-50 > EfficientNet-B0 > MobileNetV3-Small	0.6994

Table 8. Pairwise macro-F1 margins against seed noise. The margin is computed within each seed and then averaged. Pooled seed SD is the root-mean-square of the two models’ across-seed standard deviations. A margin comfortably exceeding the seed SD, with a consistent sign across all seeds, is a difference the seed alone does not explain; one below it is not claimed. With three seeds these are descriptive ratios, not significance tests.

Setting	Comparison	Seeds Mean margin Pooled seed SD Margin / seed SD Sign consis
PlantVillage in-domain	ResNet-50 - EfficientNet-B0	1 -0.0057 1/1
PlantVillage in-domain	ResNet-50 - MobileNetV3-Small	1 -0.0028 1/1
PlantVillage in-domain	EfficientNet-B0 - MobileNetV3-Small	1 0.0028 1/1
PlantVillage → PlantDoc Core	ResNet-50 - EfficientNet-B0	1 0.0055 1/1
PlantVillage → PlantDoc Core	ResNet-50 - MobileNetV3-Small	1 0.0432 1/1
PlantVillage → PlantDoc Core	EfficientNet-B0 - MobileNetV3-Small	1 0.0376 1/1
PlantDoc Core in-domain	ResNet-50 - EfficientNet-B0	1 0.0067 1/1
PlantDoc Core in-domain	ResNet-50 - MobileNetV3-Small	1 0.1153 1/1
PlantDoc Core in-domain	EfficientNet-B0 - MobileNetV3-Small	1 0.1086 1/1

Table 9. BCa bootstrap 95% confidence intervals for accuracy and macro-F1, seed 42, 10000 resamples. Intervals describe sampling variability of the fixed evaluation set for one trained checkpoint; they are not seed variance, which is reported separately.

Setting	Model	N Accuracy Accuracy 95% CI Macro-F1 Macro-F1 95% CI
PlantVillage in-domain	ResNet-50	10709 0.9925 [0.9908, 0.9940] 0.9899 [0.9870, 0.9922]
PlantVillage in-domain	EfficientNet-B0	10709 0.9968 [0.9956, 0.9978] 0.9956 [0.9938, 0.9969]
PlantVillage in-domain	MobileNetV3-Small	10709 0.9950 [0.9935, 0.9962] 0.9927 [0.9901, 0.9948]
PlantDoc Core in-domain	ResNet-50	225 0.6933 [0.6311, 0.7511] 0.6994 [0.6526, 0.7721]
PlantDoc Core in-domain	EfficientNet-B0	225 0.6800 [0.6178, 0.7378] 0.6927 [0.6457, 0.7598]
PlantDoc Core in-domain	MobileNetV3-Small	225 0.5556 [0.4889, 0.6222] 0.5841 [0.5352, 0.6675]
PlantVillage → PlantDoc Core	ResNet-50	1951 0.2778 [0.2584, 0.2978] 0.2414 [0.2253, 0.2595]
PlantVillage → PlantDoc Core	EfficientNet-B0	1951 0.2742 [0.2553, 0.2942] 0.2359 [0.2203, 0.2538]
PlantVillage → PlantDoc Core	MobileNetV3-Small	1951 0.2419 [0.2230, 0.2609] 0.1982 [0.1831, 0.2158]

Table 10. Pairwise McNemar exact tests on per-item correctness, seed 42. ‘A only’ and ‘B only’ are the discordant counts the test is computed from. Holm-adjusted p-values control the family-wise error rate across all pairwise tests reported here.

Setting	Comparison	Acc. A Acc. B Diff.	A only B only Discordant p (exact) p (Holm)
PlantVillage in-domain	ResNet-50 vs EfficientNet-B0	0.9925 0.9968 -0.0043	18 64 82 <0.0001 <0.0001
PlantVillage in-domain	ResNet-50 vs MobileNetV3-Small	0.9925 0.9950 -0.0024	34 60 94 0.0095 0.0363
PlantVillage in-domain	EfficientNet-B0 vs MobileNetV3-Small	0.9968 0.9950 +0.0019	37 17 54 0.0091 0.0363
PlantDoc Core in-domain	ResNet-50 vs EfficientNet-B0	0.6933 0.6800 +0.0133	20 17 37 0.7428 1.0000
PlantDoc Core in-domain	ResNet-50 vs MobileNetV3-Small	0.6933 0.5556 +0.1378	48 17 65 0.0002 0.0012
PlantDoc Core in-domain	EfficientNet-B0 vs MobileNetV3-Small	0.6800 0.5556 +0.1244	47 19 66 0.0008 0.0045
PlantVillage → PlantDoc Core	ResNet-50 vs EfficientNet-B0	0.2778 0.2742 +0.0036	179 172 351 0.7488 1.0000
PlantVillage → PlantDoc Core	ResNet-50 vs MobileNetV3-Small	0.2778 0.2419 +0.0359	207 137 344 0.0002 0.0013
PlantVillage → PlantDoc Core	EfficientNet-B0 vs MobileNetV3-Small	0.2742 0.2419 +0.0323	205 142 347 0.0008 0.0045

Table 11. Stability of each pairwise verdict across independent training seeds. Each seed’s McNemar test is computed on that seed’s own checkpoints and Holm-adjusted within that seed. A comparison significant in every seed is a conclusion about the architectures; one significant in a single seed is a statement about that run.

Setting	Comparison	Seeds Significant (Holm)	Sign consistent	Mean acc. diff.
PlantVillage in-domain	ResNet-50 vs EfficientNet-B0	1 1/1	yes	-0.0043
PlantVillage in-domain	ResNet-50 vs MobileNetV3-Small	1 1/1	yes	-0.0024
PlantVillage in-domain	EfficientNet-B0 vs MobileNetV3-Small	1 1/1	yes	+0.0019
PlantDoc Core in-domain	ResNet-50 vs EfficientNet-B0	1 0/1	yes	+0.0133
PlantDoc Core in-domain	ResNet-50 vs MobileNetV3-Small	1 1/1	yes	+0.1378
PlantDoc Core in-domain	EfficientNet-B0 vs MobileNetV3-Small	1 1/1	yes	+0.1244
PlantVillage → PlantDoc Core	ResNet-50 vs EfficientNet-B0	1 0/1	yes	+0.0036
PlantVillage → PlantDoc Core	ResNet-50 vs MobileNetV3-Small	1 1/1	yes	+0.0359
PlantVillage → PlantDoc Core	EfficientNet-B0 vs MobileNetV3-Small	1 1/1	yes	+0.0323

3 Efficiency

4 Complete Calibration and Selective Prediction

5 Ranking Stability Across Seeds

6 Statistical Tests

7 Training Curves

Every seed is drawn rather than averaged. Early stopping fires at different epochs, so a mean curve over epoch index would silently truncate to the earliest stop and misrepresent the runs that trained longer.

8 Risk–Coverage and Seed Spread

9 Per-Run Diagnostics

Confusion matrices and reliability diagrams are generated for every run and evaluation. They are released as individual files rather than reproduced here, since the full set runs to dozens of figures:

- experiments/ica26/figures/fig_confusion_<run>_<eval>.pdf
- experiments/ica26/figures/fig_reliability_<run>.pdf

10 Per-Seed Values

Every aggregated table above has a per-seed companion in CSV, so each reported mean and standard deviation can be recomputed without rerunning any experiment:

- table2_in_domain_performance_per_seed.csv
- table3_cross_domain_performance_per_seed.csv
- table4_efficiency_per_seed.csv
- table5_domain_shift_degradation_per_seed.csv
- table6_calibration_per_seed.csv

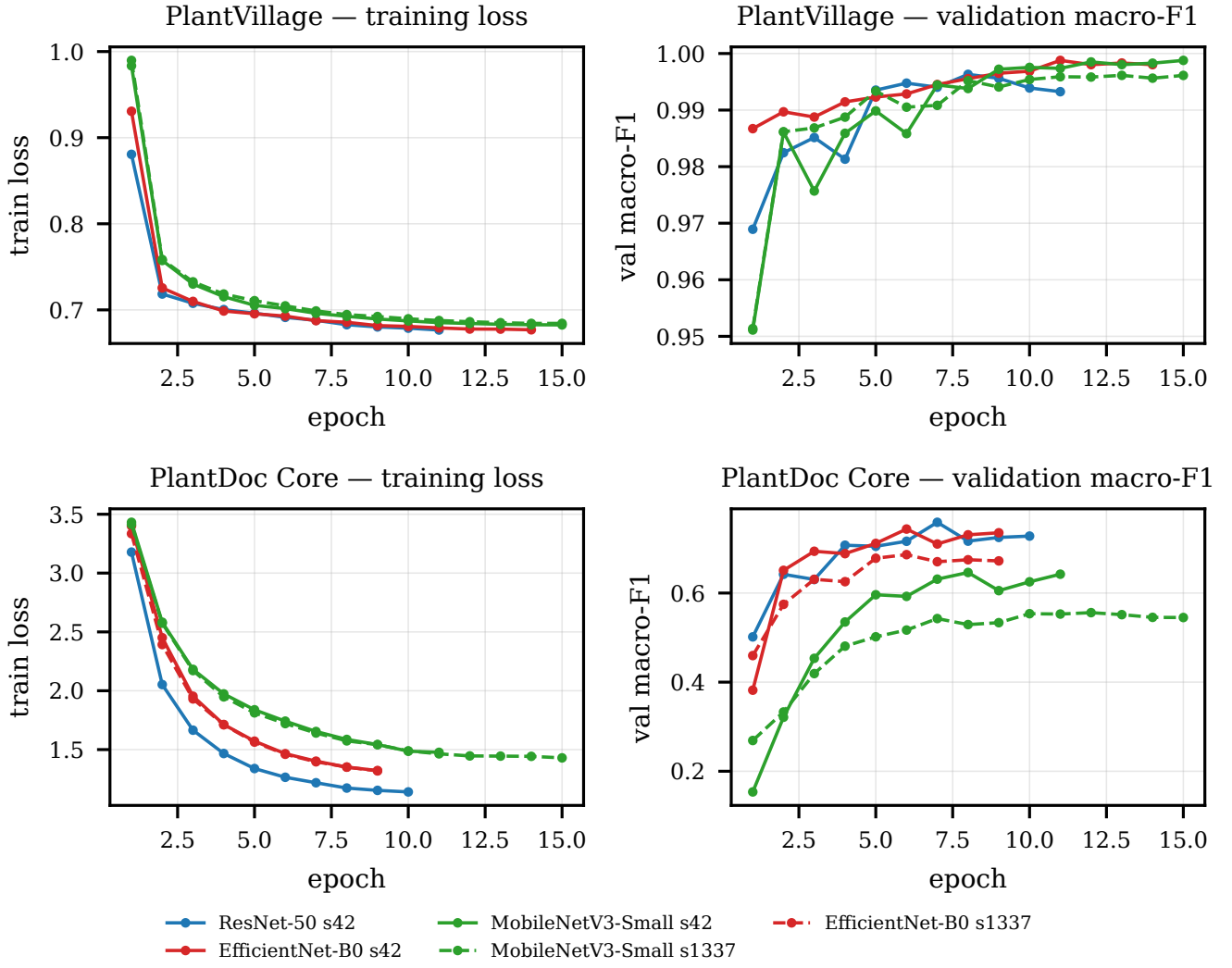


Fig. 1. Training loss and validation macro-F1. Colour is the backbone, dash pattern is the seed.

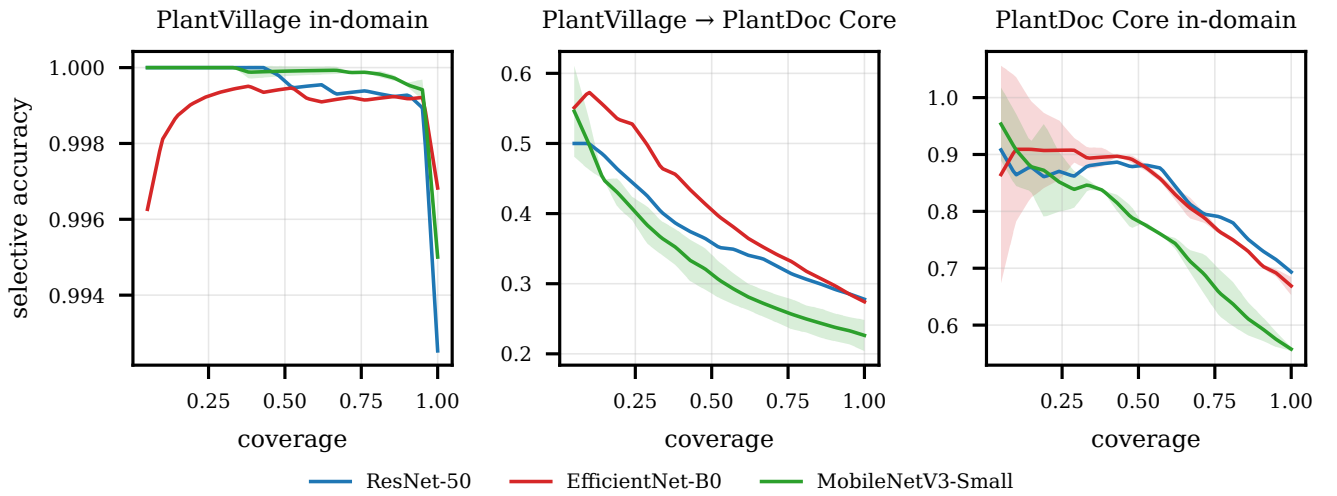


Fig. 2. Confidence-abstention curves, seed-averaged on a common coverage grid with a $\pm$ one standard deviation band.

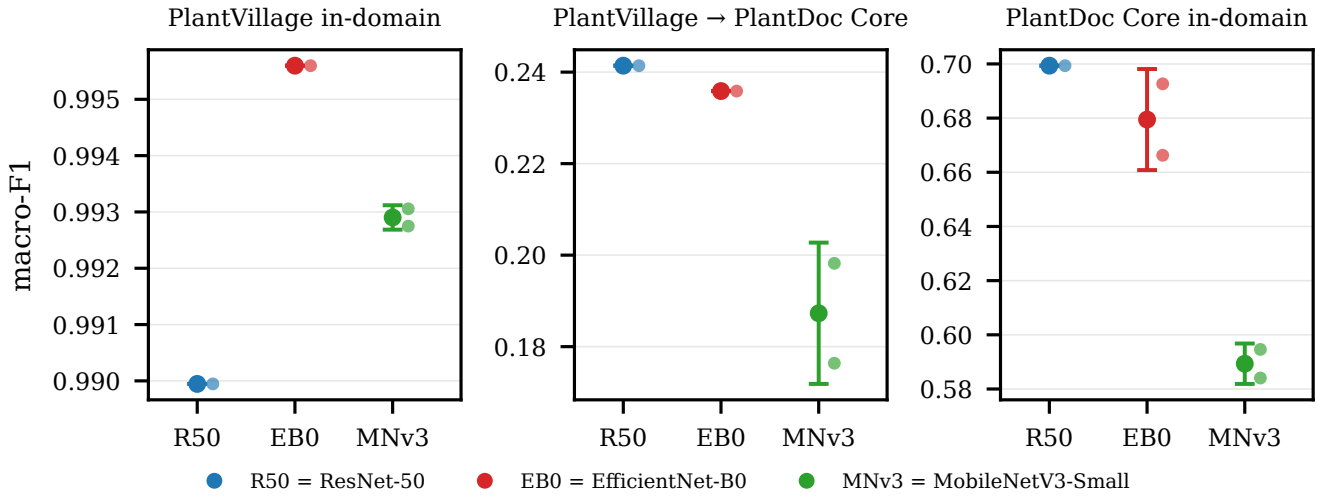


Fig. 3. Backbone macro-F1 by evaluation setting: mean with a seed standard deviation bar, and each individual seed drawn beside it. With three seeds the individual points carry more information than the interval computed from them.